mlX 2.0 Regression - Lab 2

Final Report

Index No : 230601B

1 Competition

The goal of this competition is to predict a song's popularity score (0-100) using real-world music data from audio features and artist stats to track metadata. Different techniques of machine learning are used to analyze trends in danceability, energy, valence, and more to train a model that forecasts how listeners will react before a song even hits the charts.
The Challenge: Predict continuous popularity scores (not just hits/flops).

2 Dataset and Preprocessing

The dataset includes real Billboard-tracked tracks Each row represents a single music release with its audio features and derived characteristics.

2.1 Data Cleaning

Duplicate rows were removed from the training dataset. Several high-correlated columns(`emotional_charge_1`, `emotional_resonance_2`, `emotional_charge_0`)or low-value columns, such as `track_identifier`, and composition-related labels were dropped to reduce noise and dimensionality.

2.2 Handling Missing Values

Numerical features were imputed using mean values computed from the training set. Categorical features were filled using the most frequent category (mode). The same statistics were applied to the test set to maintain consistency.

2.3 Date Feature Engineering

The `publication_timestamp` column was converted into datetime format. From this, new features such as year, month, day, and day of the week were extracted. The original timestamp column was then removed.

2.4 Feature Engineering

For grouped features (e.g., `duration_0, duration_1, duration_2`), two new features were created:

- Average value (`base_avg`)
- Variance (`base_var`)

Additionally, interaction features such as:

- `duration` × `intensity`

- `texture` × `groove`

were introduced to capture relationships between variables.

The final dataset contained 83 features, including 80 numerical and 3 categorical variables.

3 Approaches

3.1 Linear Regression (Baseline)

A linear regression model was used as a baseline due to its simplicity and interpretability. Numerical features were scaled, and categorical features were one-hot encoded. The model works by fitting a linear equation that minimizes the squared difference between predicted and actual values using ordinary least squares.

Advantages:

- Fast to train
- Easy to interpret
- Provides a performance baseline

Drawbacks:

- Cannot capture non-linear relationships effectively
- Requires manual feature engineering for interactions

3.2 XGBoost Regression

XGBoost was selected due to its strong performance on tabular data and ability to model complex non-linear relationships. It works by building an ensemble of decision trees sequentially, where each new tree corrects the errors of the previous ones using gradient boosting.

Hyperparameters such as tree depth, learning rate, and regularization terms were tuned to improve performance.

Advantages:

- Captures non-linear patterns
- Built-in regularization reduces overfitting
- High predictive accuracy

Drawbacks:

- More complex to tune
- Less interpretable than linear models

3.3 CatBoost

CatBoost was also tested as it handles categorical features natively without requiring one-hot encoding. It uses an ordered boosting approach and target-based encoding to process categorical variables while reducing prediction shift and overfitting.

Advantages:

- Handles categorical data efficiently
- Reduces preprocessing effort

Drawbacks:

- Slower training time
- Performance depends on tuning budget

4 Evaluation Metrics

Although RMSE was the primary competition metric, additional metrics were used for a more comprehensive evaluation:

- **RMSE**: Measures large errors more heavily and is the main competition metric. It is sensitive to outliers, meaning models with occasional large mistakes will be penalized more strongly.
- **MAE**: Represents average prediction error in understandable units. Unlike RMSE, it treats all errors equally, making it more robust to outliers and useful for interpreting typical model performance.
- $\mathbf{R^2}$: Indicates how much variance in the target is explained by the model. Higher values suggest better explanatory power, though it does not directly reflect prediction error and can sometimes be misleading for non-linear patterns.

5 Results and Comparison

Model	RMSE	MAE	R^2
Linear Regression	19.49	15.76	0.19
CatBoost	12.91	9.26	0.64
XGBoost	**10.50**	**6.21**	**0.76**

Table 1: Validation Performance Comparison

Linear regression showed high error and low explanatory power, indicating underfitting. This suggests that the underlying relationships in the data are not purely linear and require more expressive models to capture important feature interactions.

Both tree-based models significantly improved performance by capturing non-linear relationships and interactions automatically. CatBoost achieved a strong improvement over the baseline, particularly in RMSE and MAE, demonstrating the benefit of handling categorical features more effectively.

XGBoost achieved the best balance between bias and variance, resulting in the lowest RMSE and highest R^2. The substantial reduction in MAE also indicates that its predictions are consistently closer to the true values, not just on average but across most samples.So that proves the best model tested to be as xgboost regression.

Final Results:

- Public Score: 9.8897
- Private Score: 9.7768

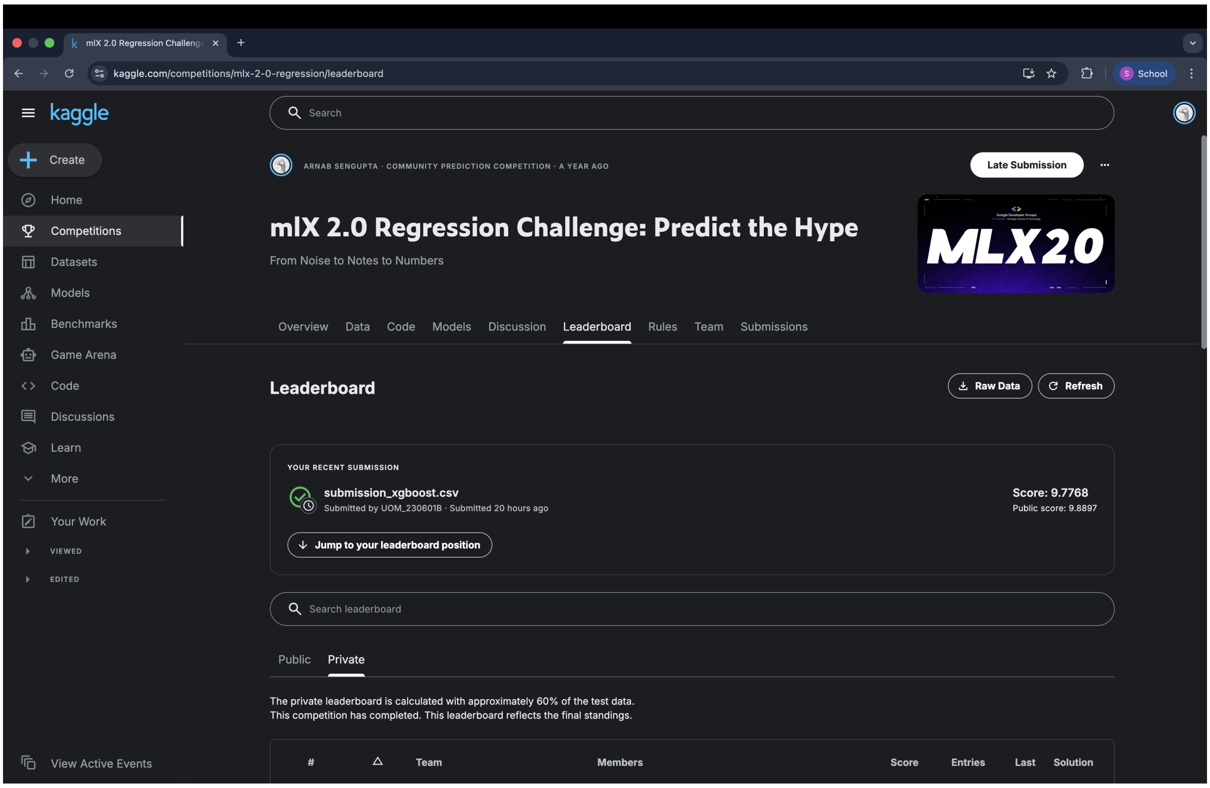

Figure 1: Leaderboard

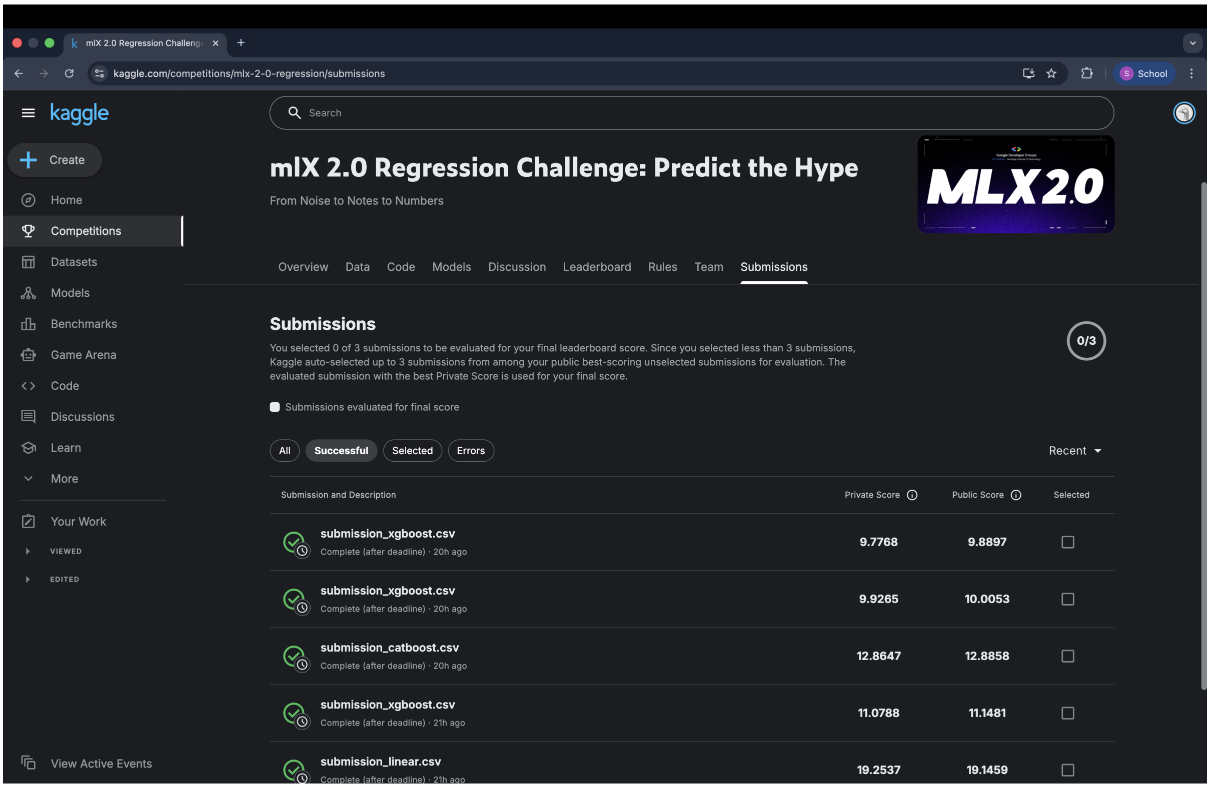

Figure 2: Submissions